

A Packing of 8 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 8 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.742631. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.699736, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 8 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.742631 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 8$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.699736, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 8/7 = 2.742857$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N8.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
137131583141903227063261163165898947759360410386144332868297247133694175
122301391692232748558340175100337451134635436712191021196777020095675999
2901863096044913987714794195093495211379652319999
```

denominator:

```
4999999999999489009429091522020266243746635219208371491007239454945250194
486762168220600859700483134641371402393520702625083379379309013601801392
600025012306246017977516456920593443238323861368
```

B Coordinates

8 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N8.txt`.

```
63110121956 702641142873 296322763419 -124458108421176366207535 654815404661 -210992619739 -428444806118854050055 -478368123242 -54469100756 289336539683
251355191487 120071876298 -678962547308 -5151300347263142474082 109005754311 110191682279 -9009877197479881998205 -195497460632 95245047864 29184138926
61534802241 -66901295984 250000666778 37555501276 -131699356323 -401967446155 -170562809017 -629457989925 10666633 -378894696502 104244901529 322722005664
-222755347053 214444681448 105483980656 -1264577557482642415166 -228606687613 -257278079097 20439225087541296680036 -423906228643 -66580112925 -192505839881
-594519118785 75921687078 -176241343165 87441047990 428073278518 -116066492941 -87904368452 27734924248317576009192 115572541092 -342918572680 339863320855
-438955749696 38163775367 -435769544081 61437758238 178201676245 -116391795498 454215343420 1536500273399163216259 -360219636719 64963575849 -428649976238
-422210382668 -380982780280 123137245672 -2319664899892533515248 -225677877403 171713834422 43887776803613874910876 9524204499 445654718105 336212510715
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343463034534 -89515325569 -261358472774 150546741832472779382039 375451653418 -20817960635 31559009587621126876661 -256243872579 -443490618247 -73671712263
-6345554645 455107942668 520728230168 143974051823 -219152333557 381386238410 95975383606 -619993697052396533832 -113819298806 -369623349198 -158920457508
-209473761378 306782456571 443128550556 51979777757101103130654 435547196087 -509713145816 263575953855670328573 144091387965 -27950573360 -479556701458
-176755086432 153496995084 -255734717550 -193815952630079999350 525507027140 14306195958 223077449499 -245138261487 399489598720 106422740827 -19635172456
120468881961 -193459667309 -59113889450 -3945174997316985532243 287994090077 100278524764 89101104514 164824264972 130890932295 70860591222 274100005211
292135899476 76358145774 -81285155000 835214517371 -237022701823 -182756730603 520909335916 6854466991263096909610 -705795294533 -12366485431 -89805887283
214794496898 -226530300594 473023092029 -19667703474134578721358 495291443872 -312370182057 664096337144712085064 318095542337 -554378580291 109949080700
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475039870781 141331098063 -156134474136 -342399364228854027830 -515990346106 -409127501995 31409440927390854320326 390579122568 -352506295961 -129446565122
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-13674104079 -133598990425 116504520761 703209637308258853859 -357995099014 337260472627 2492096149426736661613 -337082153595 -205538077210 -388461474245
-725774805977 27964892460 -243794575026 -414840676485042097691 138612483770 397033024004 15189027266657079594050 -237014942888 355292754821 551374851083
242828191251 -351012052439 576200762407 -6081897503036051658683 -85112124643 448132891953 264325463851-227507284483 152382297461 501160192807 -370292424809
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-29288550792 373626258028 262603689683 -3354208189776637866642 -503916977399 -91688315777 -3172438412559799347431 -278693339423 -149086913114 -118993192172
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```